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MDM
MASTER DATA MANAGEMENT



MAASTRICHT DATA SCIENCE PROGRAM

DHL

WHAT?

- **5 month University level Data Science course**, covering an introduction to all relevant data science topics
- The program is offered **online** and will consist of virtual meetings and independent learning in order to master all the topics that are discussed.

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- Participant will be taught, guided **and supervised by OU data science professors and supported by tutors.** Director of the course is **Pr. Dr. Stefano Bromuri,** who also works with DHL on a part time basis to research best approaches for our data science projects
- Working language is **English**
- A **certification** can be obtained after the successful completion of a number of tests.
- **The Professional Program diploma** obtained counts for 10 ECTS points and can count towards a **Bachelor or Master degree**

WHAT? - 9 MODULES

Scientific Programming I and II Roadmap, introduction to CRISP-DM methodology. Importance of programming in Data Science. Fundamental concepts of programming and introduction to Python.

Databases, Data Formats, Data Transfer Relational databases basics using SQL. Manipulate data and build queries across multiple tables. and On-Line Analytical Processing in relational databases. Json, MongoD**Data**

Understanding and data Preparation Overview on data preparations techniques, (data cleaning, data binning, missing data, data inconsistencies), data structuring, data summarization, data statistics and basic reporting tools, hypothesis testing.

Data modelling using Machine Learning I, II, III Canonical data mining tasks, data mining process, supervised and unsupervised data mining techniques, generalization of models, over fitting and model performance.

Data Modelling for Text Analytics I and II Extracting information out of unstructured text, major techniques for analyzing and mining text.

Data Visualization in Python I and II Fundamentals of data visualization and practice communicating with data. Design principles, human perception, color theory, and effective storytelling to encode and present data to an audience once an insight has been found. Matplotlib and Seaborn.

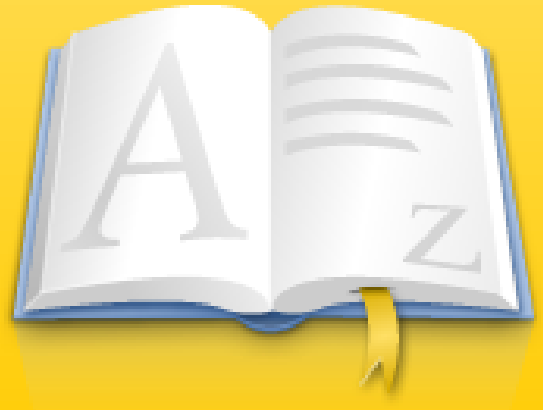
Time series analysis : Data preparation and analysis, hypothesis testing, ARIMA and SARIMAX models

Deep Learning Within the deep learning module, we will go into detail on topics such as Backpropagation, Tensorflow + Keras, LSTMs, NLP, Convolutional Neural Networks for Text and Images.

Big Data Analytics Introduction to Big Data Analytics technologies in Python.

WHEN – PERSONAL INVESTMENT





- Look and feel session – t minus 3 weeks
- Small entrance exam – t minus 2 weeks
- Program will take 5 months
- Lectures will take place on Tuesdays and Thursdays between 12 and 14 h CET (some exceptions)
- Every 4 weeks, a lab session with exercises will be organized.
- Pre-recorded videos with exercises will be made available. Each lecture is recorded and can be viewed afterwards
- You are required to attend at least 50% of the classes
- The total study time required will be of 280 hours
- Each hour of lecture → 2-3 hours of study
- You will need to hand in 2 main tasks throughout the course of the training

PREPARATION FOR ENTRANCE TEST

Objective

- The entrance test is held to determine the pace of the lectures and level of support required by the participants

Things to study

- Python: Accessing the environment, assigning variables, conditions, loops, being able to write a function
- SQL: Master selections, joins, clauses, update, order by, insert statements
- Refreshing basic Statistics helps

Tutorials

- SQL : study until ADVANCE- Using Joins : https://www.tutorialspoint.com/sql/index.htm#
- Python : study until BASIC – exceptions : https://www.tutorialspoint.com/python3/index.htm

Test

- Test will be made available to you via a link approx. **3 weeks before the start** of your term.
- You will have **2 weeks** to complete the entrance test.
- An **average of 10 hours** is needed to complete the set of given questions.
- From the moment you start the test, you have **20 hours to finish**.
- You can try some sample questions here : https://app.testdome.com/t?GeneratorId=46
- The test is **“open book”** so you can look everything up on the internet , which is common practice